

DS-5343

WHY: Abstracting the TASK

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Índice



1. Abstract the tasks (why)
2. Define the task's verb (action)
3. Define the task's noun (objectives or elements of interest)
4. Best practices in task definition
5. Derivation of new data



“Affordances”

the design aspects of an object which suggest how the object should be used—a visual cue to its function and use.

They can be patterns
...or metaphors
...or just labels.



Car seat adjuster:
Which part of the control
affects which part of
the seat?



Logos:

What do you expect if you click on a logo?



affordance

All

Books

Images

News

Videos

More

About 2,970,000 results (0.35 seconds)

Affordances are clues about how an object should be used, typically provided by the object itself or its context. For example, even if you've never seen a coffee mug before, its use is fairly natural. The handle is shaped for easy grasping and the vessel has a large opening at the top with an empty well inside.

[What Are Affordances in Web Design? - Treehouse B](https://blog.teamtreehouse.com/affordances-web-design)
<https://blog.teamtreehouse.com/affordances-web-design>

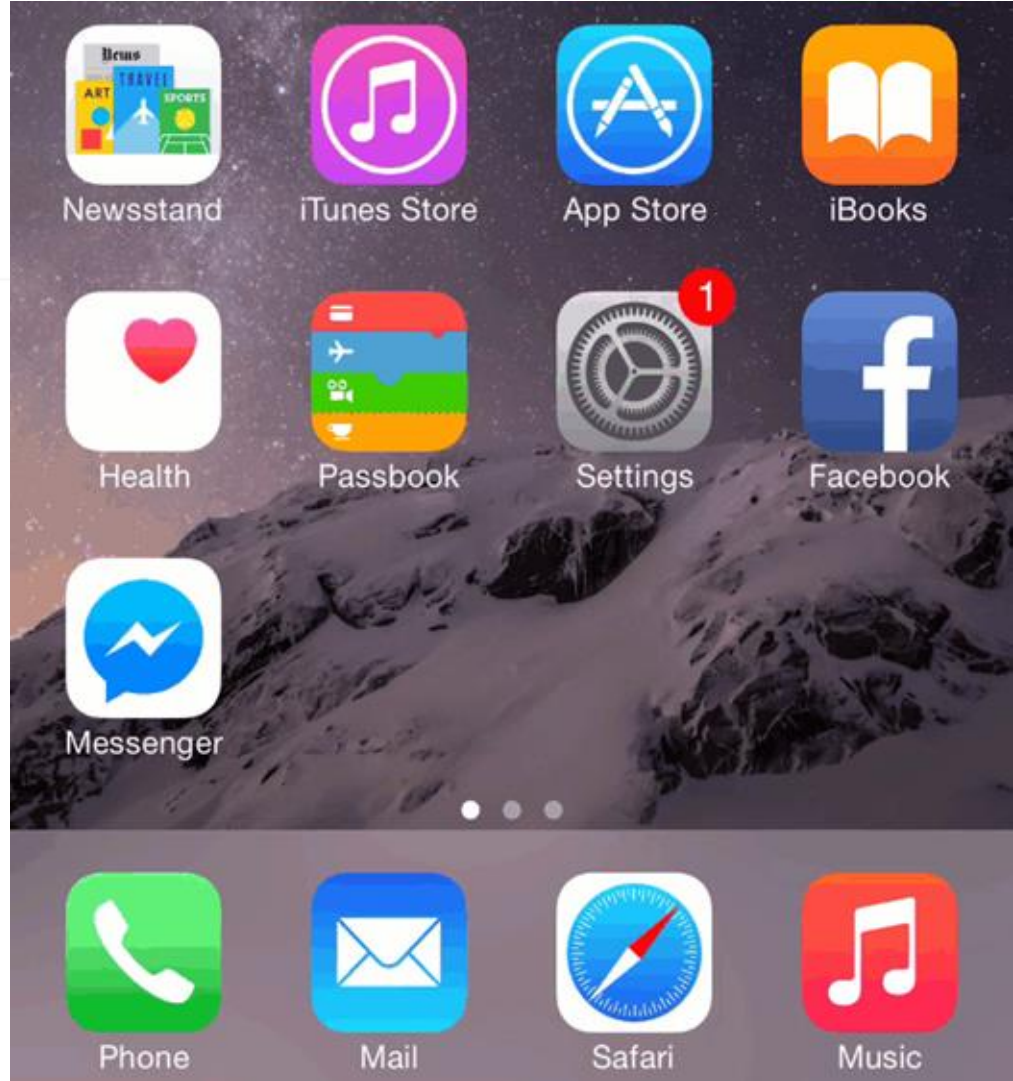
People also ask

What Affordances means?

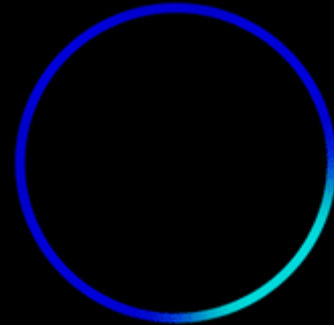
What are Affordances in psychology?

What is Affordance in UX?

iPhone:
“Wiggle” hints at what
affordance?



Alexa:
What do we guess is happening?



Counter-example: "Norman Door"



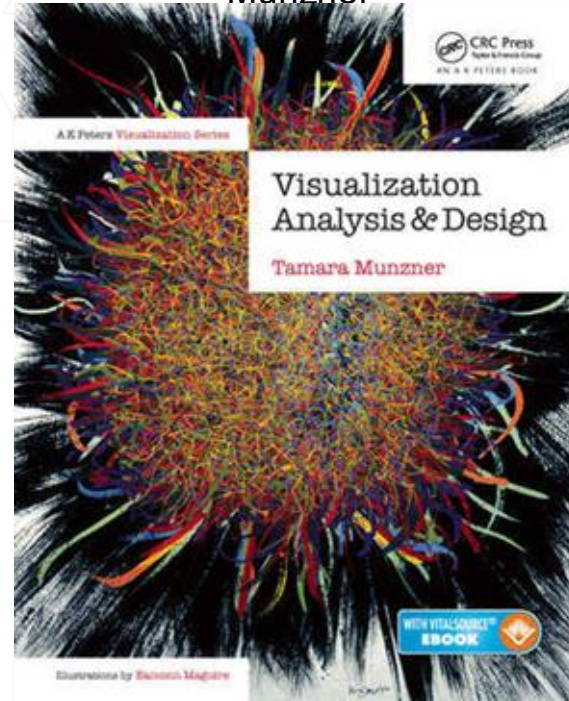
Goals of the Session

In this session, we will address **task abstraction** in data visualization, focusing on **understanding the purpose (the “why”)** behind each task. We will explore **how to define the verb (action) and the noun (objective or element of interest)** within a task. In addition, best practices for structuring tasks effectively will be presented, and methods **for deriving new data** from existing datasets will be analyzed, enabling a clearer understanding and a more effective visual representation of the information.



Book

Visualization Analysis and Design, Tamara Munzner



Chapter 3

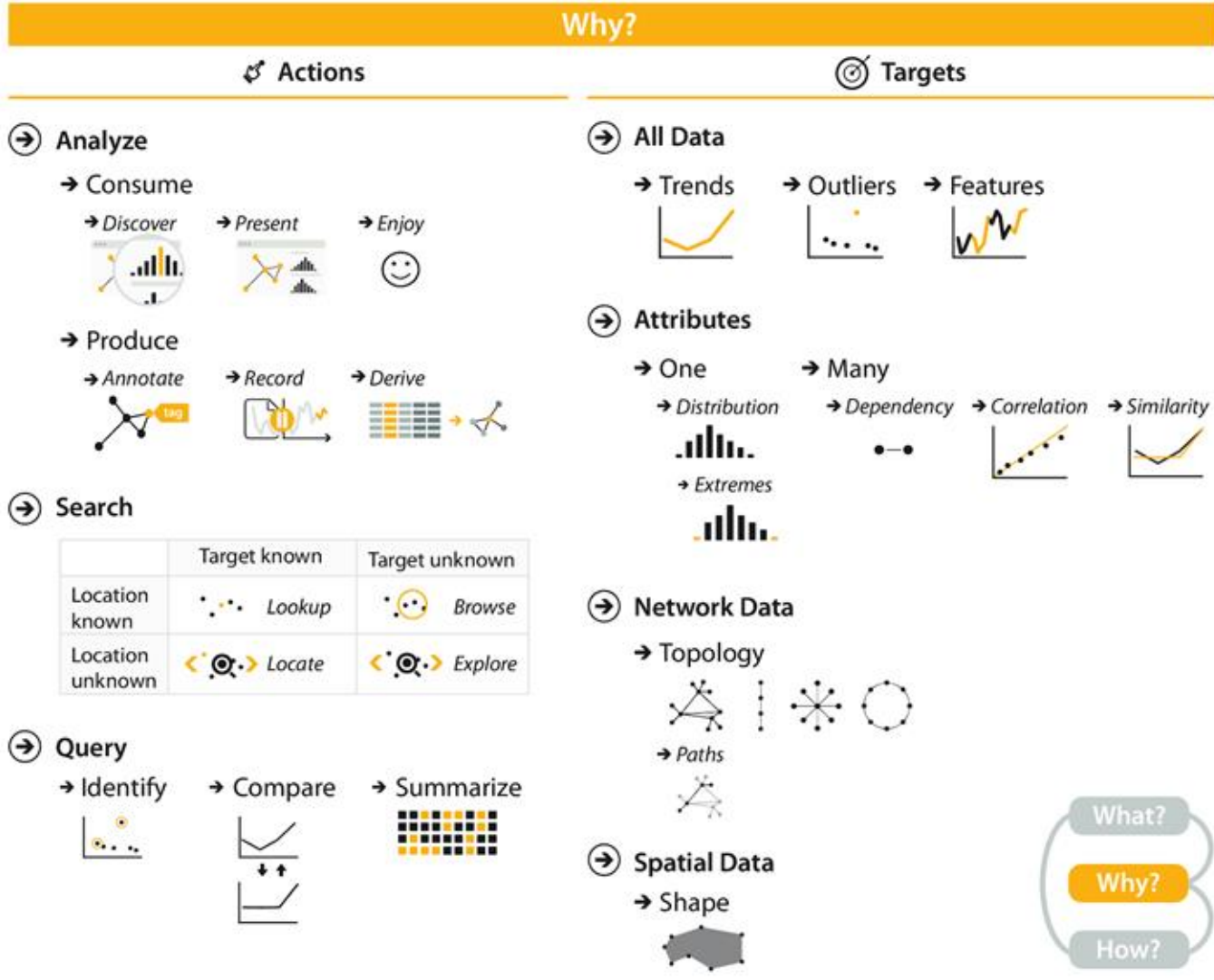
1.

Abstracting the tasks (Why)





- {action, target} pairs
 - *discover distribution*
 - *compare trends*
 - *locate outliers*
 - *browse topology*



Why analyze tasks?

- Abstract tasks are domain-independent
- Two different domain problems 👉 same task abstraction 👉 same solution
- Visualization idioms are good for some tasks and bad for others

Task construction

Action (verb)

Actions

→ Analyze

→ Consume

→ Discover



→ Present



→ Enjoy



→ Produce

→ Annotate



→ Record



→ Derive



→ Search

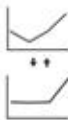
	Target known	Target unknown
Location known	• • • • Lookup	• • • • Browse
Location unknown	• • • • Locate	• • • • Explore

→ Query

→ Identify



→ Compare



→ Summarize



Target (noun)

Targets

→ All Data

→ Trends



→ Outliers



→ Features



→ Attributes

→ One

→ Distribution



→ Extremes



→ Many

→ Dependency



→ Correlation



→ Similarity



→ Network Data

→ Topology



→ Paths



→ Spatial Data

→ Shape



2.

Abstracción de Acciones



Actions verbs

Actions: Levels

Actions

→ Analyze

→ Consume

→ Discover



→ Present



→ Enjoy



→ Produce

→ Annotate



→ Record



→ Derive



→ Search

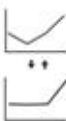
	Target known	Target unknown
Location known	 Lookup	 Browse
Location unknown	 Locate	 Explore

→ Query

→ Identify



→ Compare



→ Summarize



Three levels:

- Analyze
- Search
- Query

Try being specific

Actions: Analyze, High-level

- Consume
 - Discover vs. present
 - Enjoy
- Produce
 - Annotate, record
 - Derive

→ Analyze

→ Consume

→ Discover



→ Present



→ Enjoy



→ Produce

→ Annotate



→ Record



→ Derive



<https://distill.pub/2018/building-blocks/>

<https://www.nytimes.com/interactive/2017/08/09/upshot/game-of-thrones-chart.html>

<https://distill.pub/>

Actions: Search, Mid-level

→ Search

	Target known	Target unknown
Location known	 <i>Lookup</i>	 <i>Browse</i>
Location unknown	 <i>Locate</i>	 <i>Explore</i>

- **Target**
 - I know what I'm looking for
 - E.g., what's the salary of the CEO? 🖱️ Target known
 - E.g., who has the highest salary? 🖱️ Target unknown

ACCIONES: BUSCAR, MID-LEVEL

→ Search

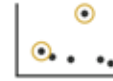
	Target known	Target unknown
Location known	 <i>Lookup</i>	 <i>Browse</i>
Location unknown	 <i>Locate</i>	 <i>Explore</i>

- LOCATION
 - I know where the mark is going to be located in the chart
 - Scales ordered alphabetically

Actions: QUERY, LOW-LEVEL

→ Query

→ Identify



→ Compare



→ Summarize



- How much of the data matter?
 - One 👉 identify
 - Some 👉 compare
 - All 👉 summarize

3.

Targets Abstraction (Nouns)



➔ All Data

➔ Trends



➔ Outliers



➔ Features



➔ Attributes

➔ One

➔ Distribution



➔ Extremes



➔ Many

➔ Dependency



➔ Correlation



➔ Similarity



➔ Network Data

➔ Topology



➔ Paths



➔ Spatial Data

➔ Shape

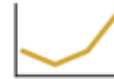


Targets: Trends, Outliers, Features

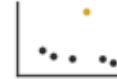
- Trends (or patterns): e.g., are sales increasing?
- Outliers: one or some that aren't like the others
- Features: 📌 task-dependent (e.g., a V-shaped recession)

➔ All Data

➔ Trends



➔ Outliers



➔ Features



One Attribute: Distributions, Extremes

→ Attributes

→ One

→ *Distribution*



→ *Extremes*



- Distribution (e.g., GDP distribution across multiple countries)
- Extremes (e.g., country with highest GDP)

Many Attributes: Dependency, Correlation, Similarity

- Dependency

(e.g., do all NBA players attended college?)

- Correlation

(e.g., do higher-paid players score more?)

- Similarity

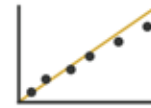
(e.g., do Curry and Hardem shoot the same?)

→ Many

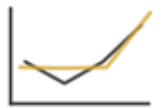
→ Dependency



→ Correlation



→ Similarity



NETWORKS: TOPOLOGY AND SHAPE

- Topology

(e.g., do all my classmates follow each other?)

- Paths

(e.g., who do I know on LinkedIn that knows Bill Gates?)

→ Network Data

→ Topology



→ Paths



Targets: Spatial data

→ Spatial Data

→ Shape



- Shape

Shape (e.g., is there a park within 100 feet of every school?)

4.

Good practices



- Be specific
 - Is your task measurable?
 - Number your tasks
 - Use Tamara's grammar
 - Describe specific attributes used
 - Don't think about the visuals
- T1: **Summarize** the stock market prices for the S&P 500 companies (Summarize, Distribution)
 - T2: **Identify** the most profitable books for the last year (Identify, Features)
 - T3: **Compare** the **trends** of COVID-19 deaths in the world (Compare, Trend)

Abstraction examples:

Policy makers in California want to test what counties had flattened the curve in the Covid 19 pandemic.

T1. Compare the (shape of the curves of the) number of positive cases per day over time between counties
(compare, trend/features).
(compare, trend/features).

Economists want to classify what type of recession was generated by the Covid 19 pandemic in different countries

T2. Compare the shape of the GDP over time per country
(compare, trend/features)

Example insights

Santa Guerra County has been the best at flattening the curve in California, with a reduction of 37% of their expected deaths by June 15, 2020.

T1. Compare the (shape of the curves of the) number of positive cases per day over time between counties
(compare, trend/features).

Example insights

- Imagine what the visualization users could be able to discover
- Made up
- Include details (names and/or numbers)
- Are very specific
- Can be tested
- Can be led to action

Task Analysis

- Who is the audience?
- What messages are most important to convey?

- Who is the audience?
 - How expert are the users?
 - How actively engaged do you expect them to be?
- What messages are most important to convey?
 - How often will your viz be read?
 - Is it meant to help solve a specific problem?



Scenario 1

- *Client:* Studio, casting directors
- *Goal:* Make the right decision
- *Format:* One-off report
- *Question:* Which actors are the best options for a new romantic comedy?



Scenario 2

- *Client:* Analytics firm
- *Goal:* Gain insight
- *Format:* Interactive feature
- *Question:* What makes for a blockbuster movie?

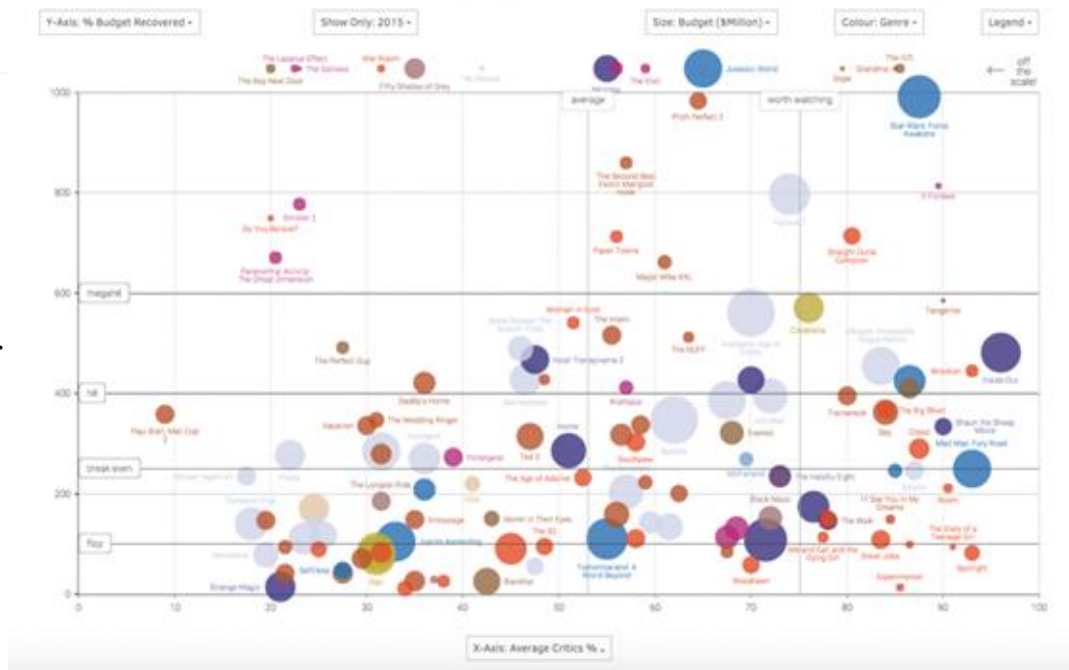
30 Actors Consistently in the Best Movies

Lead or supporting—which active actors are picking the best projects in the past decade? ¹

Rank	Actor Name	Stan Score (25) ¹	PopScore ¹
1	Paul Dano	0.3496	100
2	Leonardo DiCaprio	0.3151	97.4
3	Dominic Monaghan	0.2407	91.9
4	Brie Larson	0.2220	90.5
5	Tom Hardy	0.2106	89.6
6	Idris Elba	0.2084	89.6
7	Brendan Gleeson	0.2034	89.4
8	Philip Seymour Hoffman	0.1992	89.1
9	Michael Caine	0.1929	88.7
10	Ryan Gosling	0.1895	88.3
11	Tilda Swinton	0.1720	88
12	Carey Mulligan	0.1678	86.7
13	Jessica Chastain	0.1663	86.4
14	Mark Ruffalo	0.1577	86.3
15	Brad Pitt	0.1352	85.7
16	Jake Gyllenhaal	0.1210	84
17	Toni Collette	0.1188	82.9
18	John C. Reilly	0.1113	82.7
19	Noomi Rapace	0.1104	82.2
20	Christian Bale	0.1080	82.1
21	Benedict Cumberbatch	0.1075	81.9
22	Christopher Plummer	0.1061	81.8
23	Oscar Isaac	0.1023	81.5
24	Matt Damon	0.0733	79.3
25	Judi Dench	0.0707	79.1
26	Christoph Waltz	0.0556	77.7
27	Edward Norton	0.0425	77
28	Vera Farmiga	0.0408	76.9
29	Naomi Watts	0.0406	76.9
30	Michael Fassbender	0.0361	76.6

239	Zac Efron	-0.7241	19.7
240	Cameron Diaz	-0.7299	19.3
241	Dewayne Johnson	-0.7308	19.2
242	Ashton Kutcher	-0.7426	18.4
243	Jessica Alba	-0.7770	16.2
244	Adam Sandler	-0.7783	15.7
245	Kristen Stewart	-0.8740	8.5
246	Megan Fox	-0.8870	7.6
247	Reynold Robinson	0.0000	0.0

VS.



<https://informationisbeautiful.net/visualizations/the-hollywood-insider/>



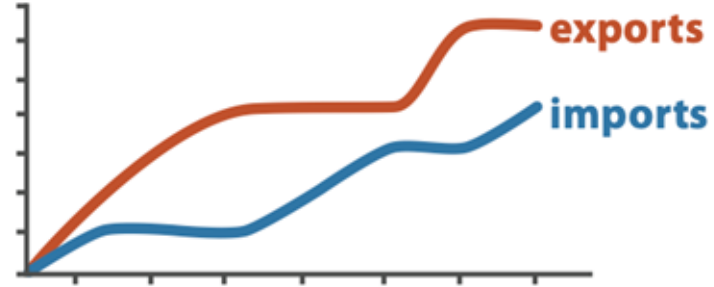
Deriving new data

→ *Derive*



TASK: Summarize the trend of the trade balance

(exports -
imports)

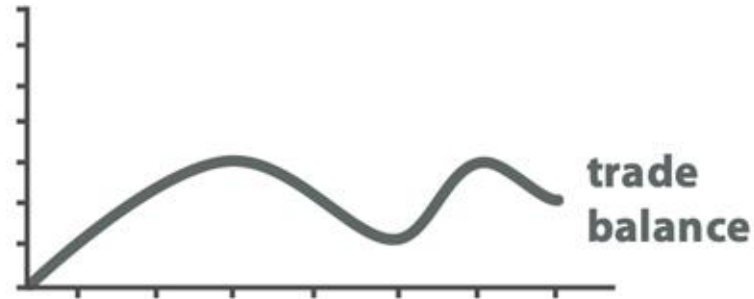


Original Data

Don't make the user compute it in their head.

TAREA: Summarize the trend of the trade balance

(exports - imports)



$$\text{trade balance} = \text{exports} - \text{imports}$$

Derived Data

Instead, display the difference.

When to derive data

- Large data? 📌 Aggregate
- Complex data? 📌 Simplify
- Dimensionality reduction
- Network/tree analytics (clusters, centralities)
- Don't make the user think!

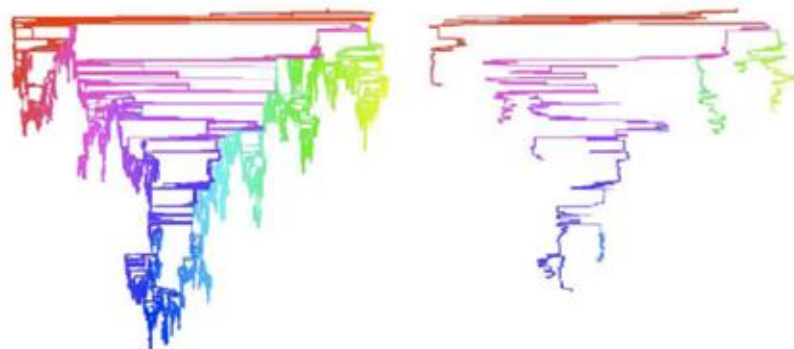
→ *Derive*



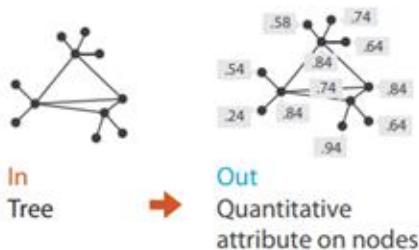
Analysis example: Derive one attribute

- Strahler number
 - centrality metric for trees/networks
 - derived quantitative attribute
 - draw top 5K of 500K for good skeleton

[Using Strahler numbers for real time visual exploration of huge graphs. Auber.
Proc. Intl. Conf. Computer Vision and Graphics, pp. 56–69, 2002.]

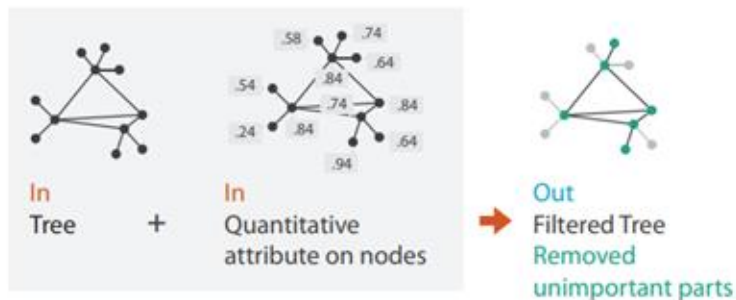


Task 1



What?	Why?
→ In Tree	→ Derive
→ Out Quantitative attribute on nodes	

Task 2

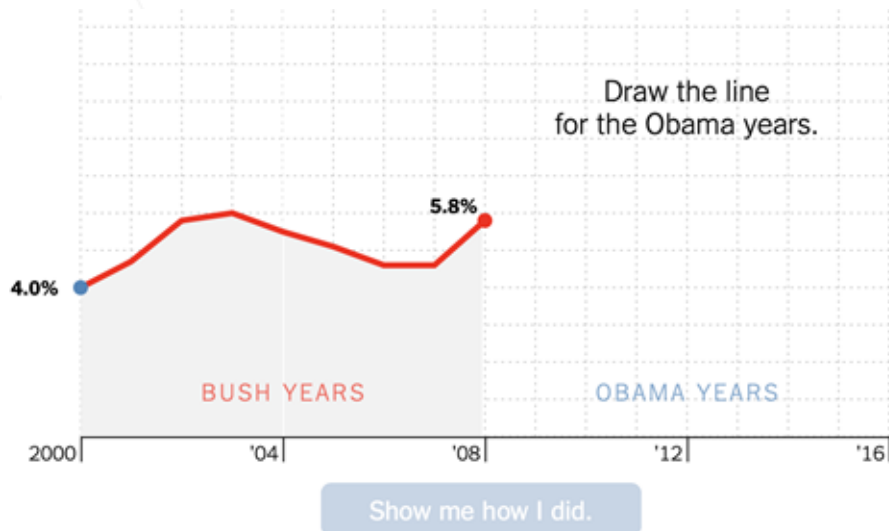


What?	Why?	How?
→ In Tree	→ Summarize	→ Reduce
→ In Quantitative attribute on nodes	→ Topology	→ Filter
→ Out Filtered Tree		

You Draw It: What Got Better or Worse During Obama's Presidency

By LARRY BUCHANAN, HAEYOUN PARK and ADAM PEARCE JAN. 15, 2017

Draw your guesses on the charts below to see if you're as smart as you think you are.



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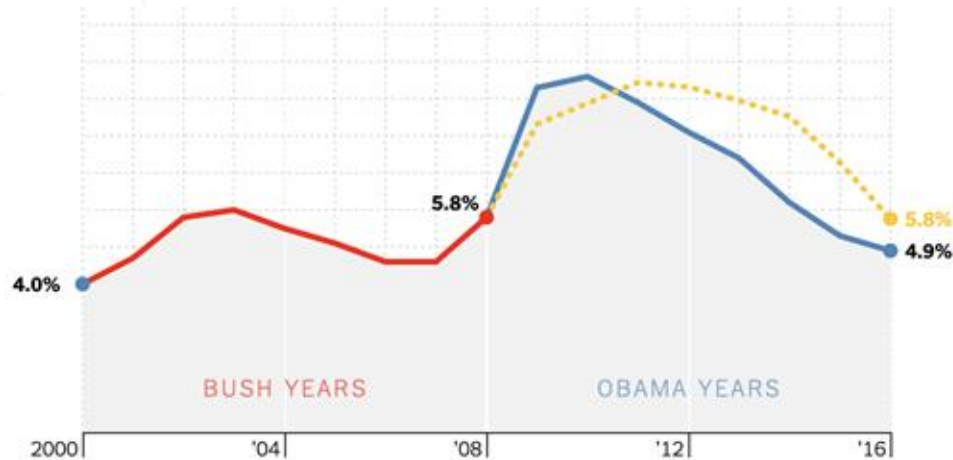
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... reached its lowest level since 2007. The current rate is a stunning decline from the 9.3 percent in 2009, the year Mr. Obama took office.

What we learned

- Abstracting the tasks (why)
- Define the verb of the task (action)
- Define the noun of the task (targets)
- Tasks good practices

From Paper to WHY (Task Abstraction)

Different problems → same abstract task → same visualization solution

Goal

Understand *what users are **trying to** do* in a visualization system.

You will:

- Read a visualization research paper
- Identify tasks (WHY)
- Translate it into Munzner's WHY framework

From Paper to WHY (Task Abstraction)

Using the **same approved paper**, identify:

- User tasks
- Task abstraction (WHY)
- Action (verb)
- Target (noun)

Select **1 research paper** (IEEE VIS, EuroVis, TVCG, etc.)

Then analyze:

1. Compare → trends
2. Identify → outliers
3. Summarize → distribution

During this class:

1. Search for a visualization research paper
2. Select **one paper**
3. **Show it to the professor for approval**



You must get approval before continuing



From Paper to WHY (Task Abstraction)

Use Munzner's categories:

- **Analyze**

- Discover / Present / Derive

- **Search**

- Lookup / Locate / Explore

- **Query**

- Identify / Compare / Summarize

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